

Complex Engineering Project (CEP)

Statistical Investigation of Chronic Kidney Disease Risk Factors and Prediction

Item Description

Item	Description
Course Title	Statistical Methods and Probability
Course Code	STA281
Semester	Spring 2026
Project Type	Complex Engineering Project (CEP)
Total Marks	20

Course Outcome

CO	Description	PO
CO1	Understand the fundamentals of statistics and probability	PO1
CO2	Analyze healthcare datasets using statistical methods and computational tools	PO1
CO3	Perform statistical inference, hypothesis testing, and interpret findings in applied contexts	PO4

Project Background

Chronic Kidney Disease (CKD) is a major global health problem that affects kidney function and can lead to severe complications if not diagnosed early. Medical indicators such as serum creatinine, blood urea, blood pressure, and hemoglobin levels are important for identifying kidney dysfunction and assessing disease progression.

In this project, students will investigate major risk factors associated with Chronic Kidney Disease and determine whether CKD occurrence can be statistically predicted using healthcare data.

Students will analyze the Chronic Kidney Disease Dataset using:

- descriptive statistics,
- probability theory,
- Bayes' theorem,
- probability distributions,
- confidence intervals,
- hypothesis testing,
- correlation analysis,
- logistic regression,
- and computational tools such as Python/Google Colab.

The project emphasizes:

- Statistical reasoning
- Healthcare analytics
- Predictive healthcare investigation
- Medical data interpretation
- Computational analysis
- Engineering problem solving

Dataset Information

Suggested Dataset

Chronic Kidney Disease Dataset

Dataset Variables

Variable	Description
Age	Patient age
BloodPressure	Blood pressure level
SpecificGravity	Urine specific gravity
Albumin	Albumin level
Sugar	Sugar level
BloodGlucoseRandom	Random blood glucose
BloodUrea	Blood urea level
SerumCreatinine	Serum creatinine level
Sodium	Sodium level
Potassium	Potassium level
Hemoglobin	Hemoglobin level
PackedCellVolume	Packed cell volume
WhiteBloodCellCount	WBC count
RedBloodCellCount	RBC count
Hypertension	Hypertension status
DiabetesMellitus	Diabetes status
CKD	0 = No CKD, 1 = CKD

General Instructions

1. Use Python or Google Colab for all analyses.
2. Include appropriate tables, graphs, and visualizations.
3. Clearly explain formulas, assumptions, and interpretations.
4. Use exact dataset variable names throughout the analysis.

5. Include healthcare interpretation for every statistical result.
6. Submit:
 - Final report
 - Python/Colab notebook
 - Presentation slides
 - Graphs and visualizations

CO–PO–WK–WP–EA Mapping Table

Task No.	Project Task Description	CO	PO / WK / WP / EA	Marks
1	Load the CKD Dataset, identify variables, perform descriptive statistics, preprocess healthcare data, detect missing values, and create visualizations using Python/Google Colab	CO1	PO1, WK2, WP1	5
2	Apply probability analysis, conditional probability, Bayes' theorem, probability distributions (Binomial, Uniform, Exponential, Normal), and the Central Limit Theorem (CLT) using the CKD dataset	CO2	PO1, WK2 WP3	5
3	Construct confidence intervals and perform hypothesis testing for healthcare variables such as SerumCreatinine and Hemoglobin with appropriate statistical interpretation	CO3	PO4, WK8, WP7	5
4	Perform correlation analysis, linear regression, healthcare investigation, visualization, and predictive interpretation using Python/Google Colab	CO3	PO4, WK6 WP7	5

SECTION 1 — Dataset Understanding, Descriptive Statistics & Visualization (Marks: 5)

Question 1

Load the Chronic Kidney Disease Dataset using Python/Google Colab and identify:

- Number of observations
- Number of variables
- Variable names
- Data types of each variable

Interpret the dataset structure and discuss the importance of healthcare data preprocessing.

Question 2

Using Python/Google Colab, perform descriptive statistical analysis for:

- BloodUrea

- SerumCreatinine
- Hemoglobin
- BloodPressure

Calculate:

1. Mean
2. Median
3. Mode
4. Variance
5. Standard deviation

Interpret each statistical measure and discuss healthcare implications.

Question 3

Identify missing or abnormal values in:

- SerumCreatinine
- BloodUrea
- Hemoglobin
- Potassium

Discuss:

- Outliers,
- inconsistent healthcare records,
- and possible effects on CKD prediction accuracy.

Question 4

Using Python/Google Colab, create and interpret:

(a) Histogram of SerumCreatinine

Analyze creatinine distribution among patients.

(b) Boxplot of BloodUrea

Identify:

- spread,
- quartiles,
- median,
- outliers.

(c) Scatter Plot of BloodPressure vs SerumCreatinine

Investigate the relationship between blood pressure and kidney function.

(d) Bar Chart of CKD Outcome

Compare CKD and non-CKD cases statistically.

Question 5

Separate the dataset into:

- CKD patients (CKD = 1)
- Non-CKD patients (CKD = 0)

Compare:

- Mean SerumCreatinine
- Mean BloodUrea
- Mean Hemoglobin
- Mean BloodPressure

Interpret healthcare findings statistically.

SECTION 2 - Probability, Probability Distribution & CLT (Marks: 5)

Question 6

Let:

- K: Patient has CKD
- C: SerumCreatinine > 1.5 mg/dL

Using the dataset, compute:

1. $P(K)$
2. $P(C)$
3. $P(K \cap C)$
4. $P(K | C)$

Interpret healthcare implications regarding kidney dysfunction and creatinine level.

Question 7

Using Bayes' theorem, estimate the probability that a patient has CKD given:

- SerumCreatinine > 1.8 mg/dL

Clearly define:

- Prior probability
- Conditional probability
- Posterior probability

Interpret the healthcare meaning of the result.

Question 8

Suppose the probability that a patient has CKD is $p=55\%$.

If 12 patients are selected randomly:

1. Find the probability that exactly 5 patients have CKD.
2. Find the probability at least 6 patients have CKD.
3. Find the probability at most 4 patients have CKD.

Use binomial distribution.

Interpret findings statistically.

Question 9

Assume SerumCreatinine follows a normal distribution with:

- Mean:
 $\mu=1.8$ mg/dL
- Standard deviation:
 $\sigma=0.6$ mg/dL

Find:

1. $P(X < 1.2)$
2. $P(1.2 < X < 2.5)$
3. $P(X > 2.5)$

Interpret:

- normal kidney function,
- moderate CKD risk,
- and severe kidney dysfunction ranges.

Question 10

Assume patient age follows a uniform distribution.

Find:

1. Probability density function (PDF)
2. $P(35 < X < 60)$
3. Mean age
4. Variance

Interpret findings statistically.

Question 11

Suppose the time interval (in hours) between consecutive CKD patient admissions follows an exponential distribution with rate parameter $\lambda=0.20$

Using exponential distribution:

1. Determine the probability density function (PDF)
2. Compute:
 $P(X>5)$
3. Determine the expected waiting time between consecutive patient admissions
4. Determine the variance of the waiting time distribution

Interpret healthcare implications regarding:

- hospital workload,
- patient flow management,
- and healthcare resource allocation.

Question 12

Suppose population **BloodUrea** level has:

- Mean:
 $\mu=55$ mg/dL
- Standard deviation:
 $\sigma=18$ mg/dL

For sample size:

$$n=36$$

Using the Central Limit Theorem (CLT), find:

1. Mean of the sampling distribution
2. Standard error
3. $P(\underline{X} > 60)$

Explain the application of CLT in kidney disease analytics and healthcare research.

SECTION 3 - Confidence Interval & Hypothesis Testing (Marks: 5)

Question 13

A sample of 64 CKD patients gives:

- Sample mean SerumCreatinine:
 $\underline{X} = 1.9$ mg/dL
- Standard deviation:
 $s = 0.7$ mg/dL

Construct a 95% confidence interval for the population mean serum creatinine level.

Interpret the interval medically.

Question 14

A healthcare researcher claims the average BloodUrea level is:
50 mg/dL

A sample of 49 patients gives:

- Sample mean:
 $\underline{X} = 57$ mg/dL
- Standard deviation:
 $s = 15$ mg/dL

Test at 5% significance level whether the average blood urea level is greater than 50 mg/dL.

Clearly state:

- Null hypothesis
- Alternative hypothesis
- Test statistic
- Critical value
- Decision
- Conclusion

Question 15

A doctor claims the average hemoglobin level is:
12.5 g/dL

A sample of 64 patients gives:

- Sample mean:
 $\bar{X}=11.4$ g/dL
- Standard deviation:
 $s=1.8$ g/dL

Test at 1% significance level whether the hemoglobin level is less than the claimed value.

Interpret healthcare implications.

SECTION 4 Correlation, Regression & Healthcare Investigation (Marks: 5)

Question 16

Using the dataset, compute the correlation between:

- SerumCreatinine and BloodUrea
- BloodPressure and SerumCreatinine
- Hemoglobin and CKD
- Potassium and Sodium

Interpret:

- strength,
- direction,
- healthcare significance.

Question 17

Perform linear regression using:

- BloodPressure → Predict SerumCreatinine level

Interpret:

- Slope
- Intercept

- Prediction equation
- Relationship significance

Discuss predictive interpretation in kidney disease analysis.

Question 18

Using Python/Google Colab, create and interpret:

1. Correlation heatmap
2. Scatter plot matrix
3. Pairplot visualization

Discuss major CKD risk factors observed from the dataset.

Question 19

Investigate whether CKD and non-CKD patients differ significantly in:

- SerumCreatinine
- BloodUrea
- Hemoglobin
- BloodPressure

Provide statistical and healthcare interpretation.

Question 20

Prepare a complete evidence-based conclusion regarding CKD prediction using:

- Statistical analysis
- Probability
- Bayes theorem
- Distribution analysis
- CLT
- Confidence interval
- Hypothesis testing
- Correlation
- Regression
- Visualization

Discuss:

- major risk factors,
- healthcare interpretation,

- predictive patterns,
- and preventive recommendations.

Suggested Report Structure

1. Introduction
2. Dataset Description
3. Descriptive Statistical Analysis
4. Visualization
5. Probability & Distribution Analysis
6. Confidence Interval & Hypothesis Testing
7. Correlation & Regression
8. Findings & Healthcare Interpretation
9. Conclusion
10. References